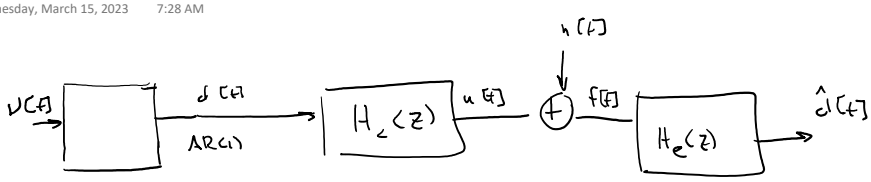


$d[t] = -.3\,d[t-1] + v[t]$

$v[t] \sim \mathcal{N}(0, 0.2)$

$H_c(z) = \dfrac{1}{1 - 0.4z^{-1}}$

$H_e(z) = h_0 + h_1 z^{-1}$

$u[t] - 0.4u[t-1] = d[t]$

$u[t] = 0.4u[t-1] + d[t]$

$\hat{d}[t] = h_0 f[t] + h_1 f[t-1]$

$f[t] = u[t] + n[t].$ $\qquad n[t] \sim \mathcal{N}(0, 0.1)$

$\langle \underline{x}, \underline{y} \rangle = E[tr[\underline{x}\,\underline{y}^H]]$

$$\begin{bmatrix} \langle \underline{y}, p_1 \rangle \\ \langle \underline{y}, p_2 \rangle \\ \vdots \\ \langle \underline{y}, p_m \rangle \end{bmatrix}$$

$\underline{y} = c_1 p_1 + c_2 p_2 + \cdots + c_m p_m + \underline{\varepsilon}$

$\underline{y} = \underbrace{[p_1 \;\; p_2 \;\; \cdots \;\; p_m]}_{A}\,\underline{c} + \underline{\varepsilon}$

$$\min_{\underline{c}} \|\underline{y} - A\underline{c}\|_2^2 = \min_{\underline{c}} \langle \underline{y} - A\underline{c}, \underline{y} - A\underline{c} \rangle$$

$$= \min_{\underline{c}} tr[E[(\underline{y} - A\underline{c})(\underline{y} - A\underline{c})^H]]$$

$$\frac{\partial}{\partial \underline{c}^*} E[tr[\underline{y}\underline{y}^H - \underline{y}\underline{c}^H A^H - A\underline{c}\underline{y}^H + A\underline{c}\,\underline{c}^H A^H]]$$

$$\frac{\partial}{\partial \underline{c}^*} tr[\underline{y}\,\underline{c}^H A^H]$$

$$\frac{\partial}{\partial \underline{c}^*} tr[A\underline{c}\,\underline{y}^H]$$

$$\frac{\partial}{\partial \underline{c}^*} tr[A\underline{c}\,\underline{c}^H A^H]$$

$\langle x, y \rangle = \langle y, x \rangle^*$

$f(t) = q(t)\,p_m(t) + r(t),$

$\deg f(t) = 2m-1$

$\deg p_m(t) = m \qquad \deg q(t) = m-1$

$\deg r(t) < \deg(p_m(t))$

$\leq m-1$

$$f(t) = q(t)\,P_m(t) + r(t),$$

$$\left[\begin{array}{l} \deg f(t) = 2m-1 \\ \deg P_m(t) = m \quad \deg q(t) = m-1 \\ \deg r(t) < \deg(P_m(t)) \\ \quad \leq m-1 \end{array}\right.$$

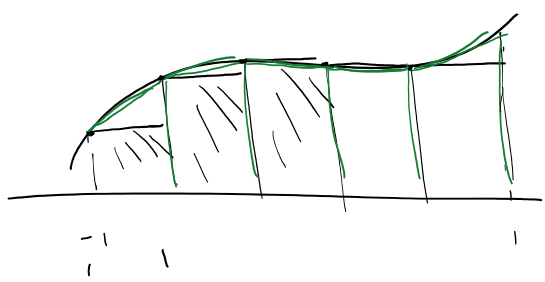
$$\int_{-1}^{1} f(t)\,dt = \int_{-1}^{1} \left(q(t)\,P_m(t) + r(t)\right) dt$$

$$= \underbrace{\langle q(t), P_m(t)\rangle}_{0} + \int_{-1}^{1} r(t)\,dt$$

$$= \int_{-1}^{1} \left(\sum_{k=0}^{m-1} \beta_k P_k(t)\right) dt$$

$$= \int_{-1}^{1} \left(\sum_{k=0}^{m-1} \beta_k P_k(t)\right) \cdot 1 \; dt$$

$$= \int_{-1}^{1} \beta_0 P_0(t) \cdot 1 \; dt$$

$P_0(t) \propto 1$

$$\left\{\begin{array}{l} \langle P_1(t), P_0(t)\rangle = 0 \\ \vdots \\ \langle P_{m-1}(t), P_0(t)\rangle = 0 \end{array}\right.$$